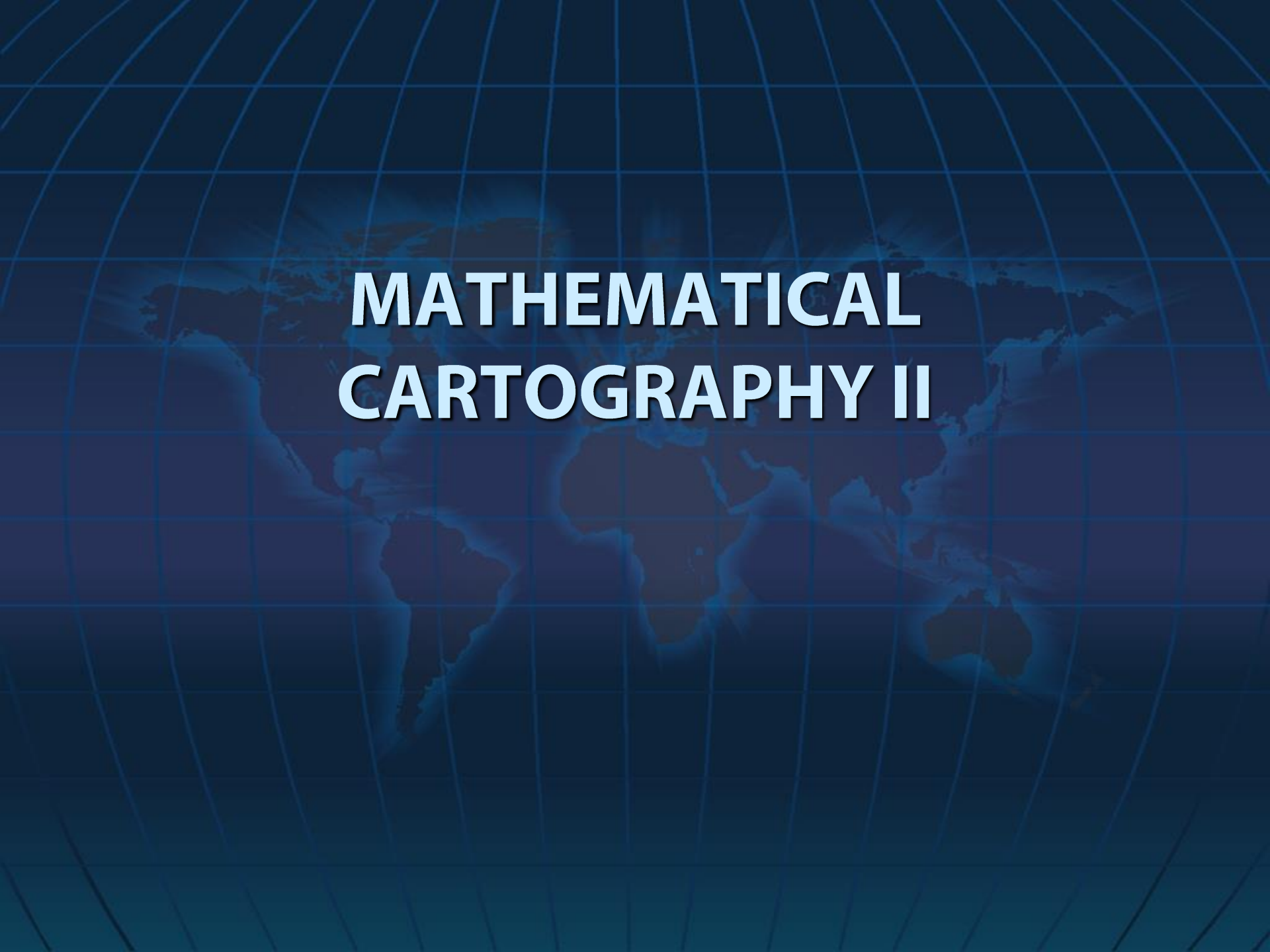


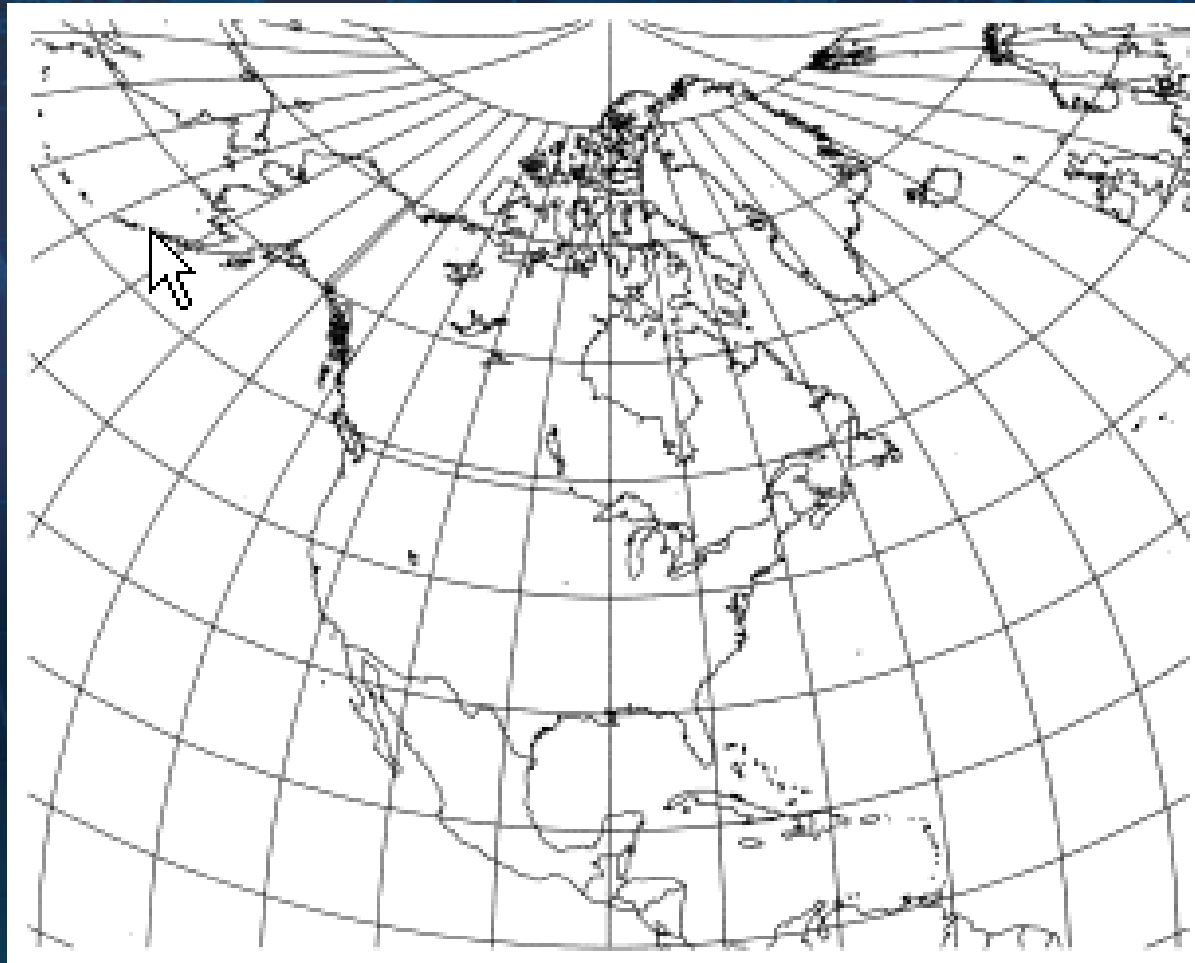
A dark blue background featuring a faint world map and a grid of latitude and longitude lines. The map is centered, showing the continents of North America, South America, Africa, Europe, Asia, and Australia. The grid lines are thin and light blue, creating a subtle pattern over the map.

MATHEMATICAL CARTOGRAPHY II

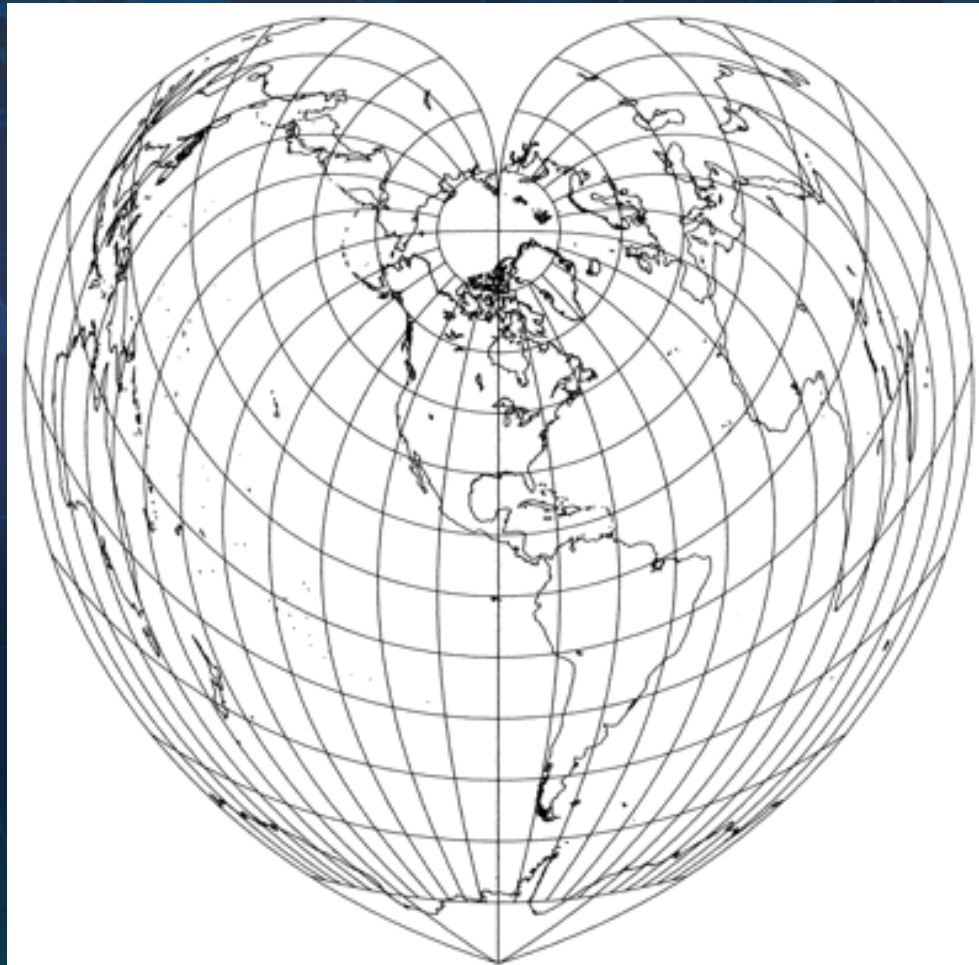
Pseudo-projections

- They preserve some properties of simple views, change others
- Conical, cylindrical, azimuthal
- The main reason for the use of these projections is to improve grid properties or eliminate distortion
- They are not conformal
- They are mainly used for world maps

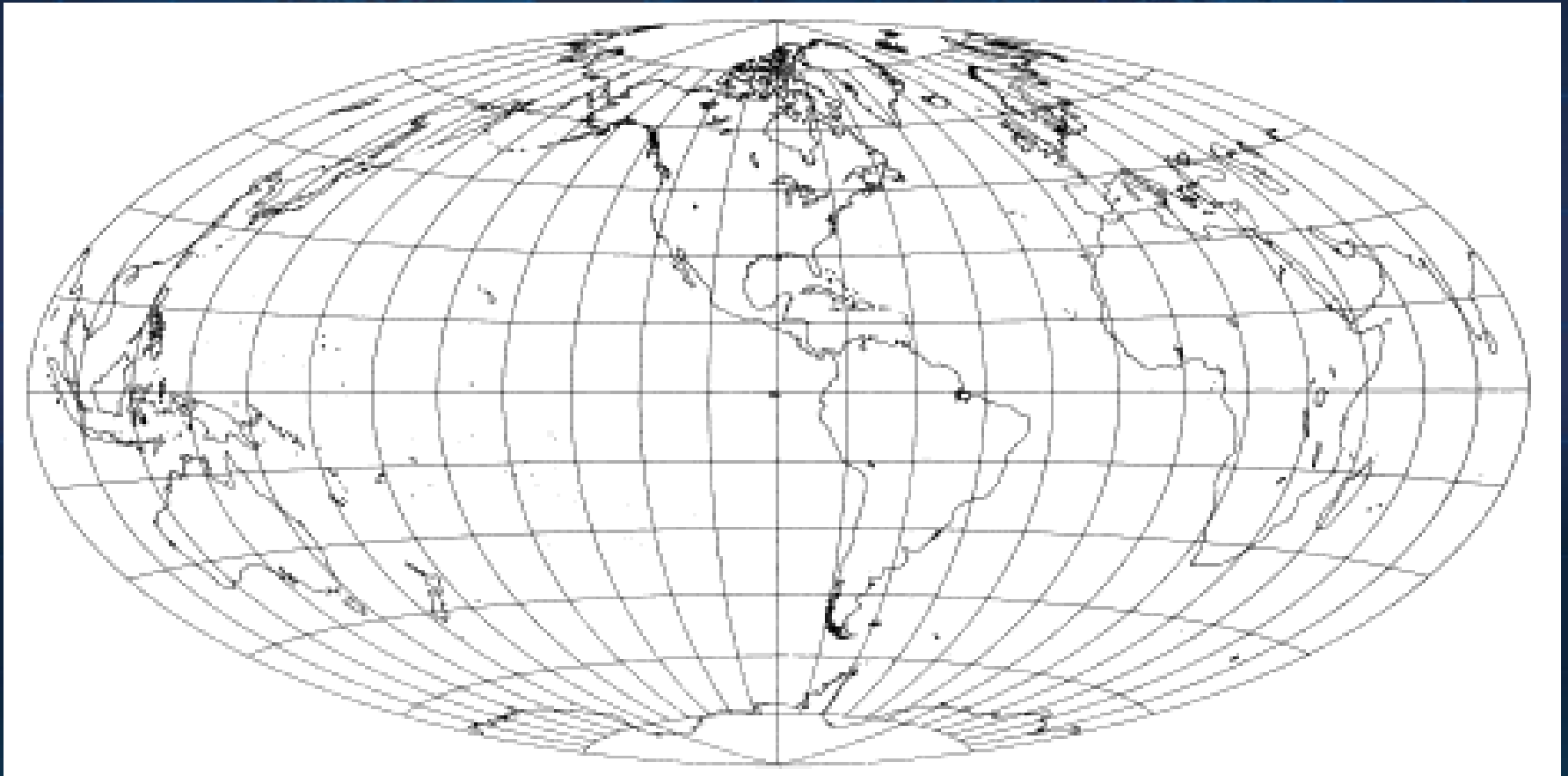
Bonne pseudo-conic projection (equivalent, equidistant in parallels)



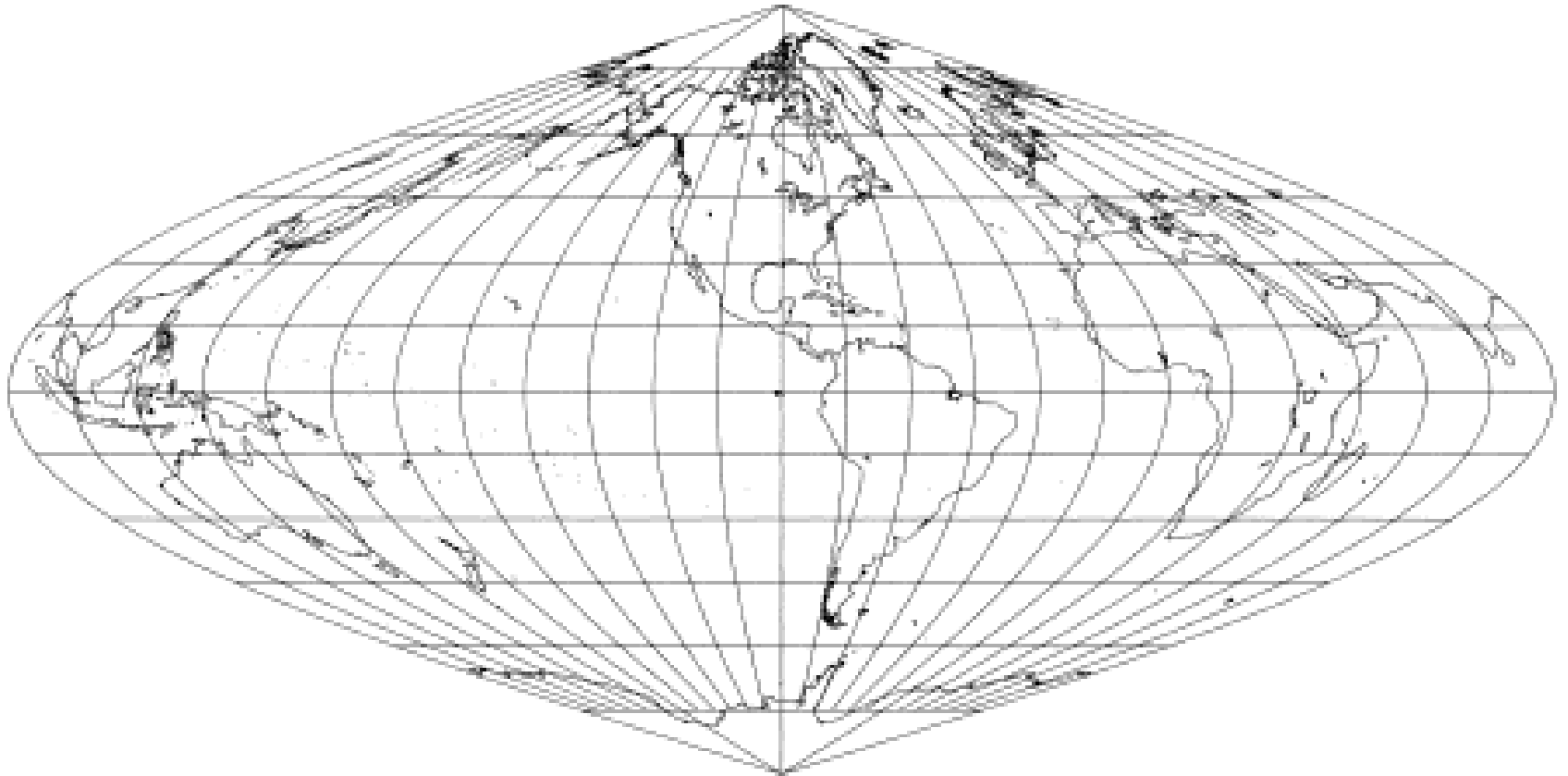
Werner-Staab pseudo-azimuthal projection (equivalent, equidistant in parallels)



Pseudo-azimuthal projection of Aitov



Pseudo-cylindrical Mercator-Sanson projection (equidistant in parallels)



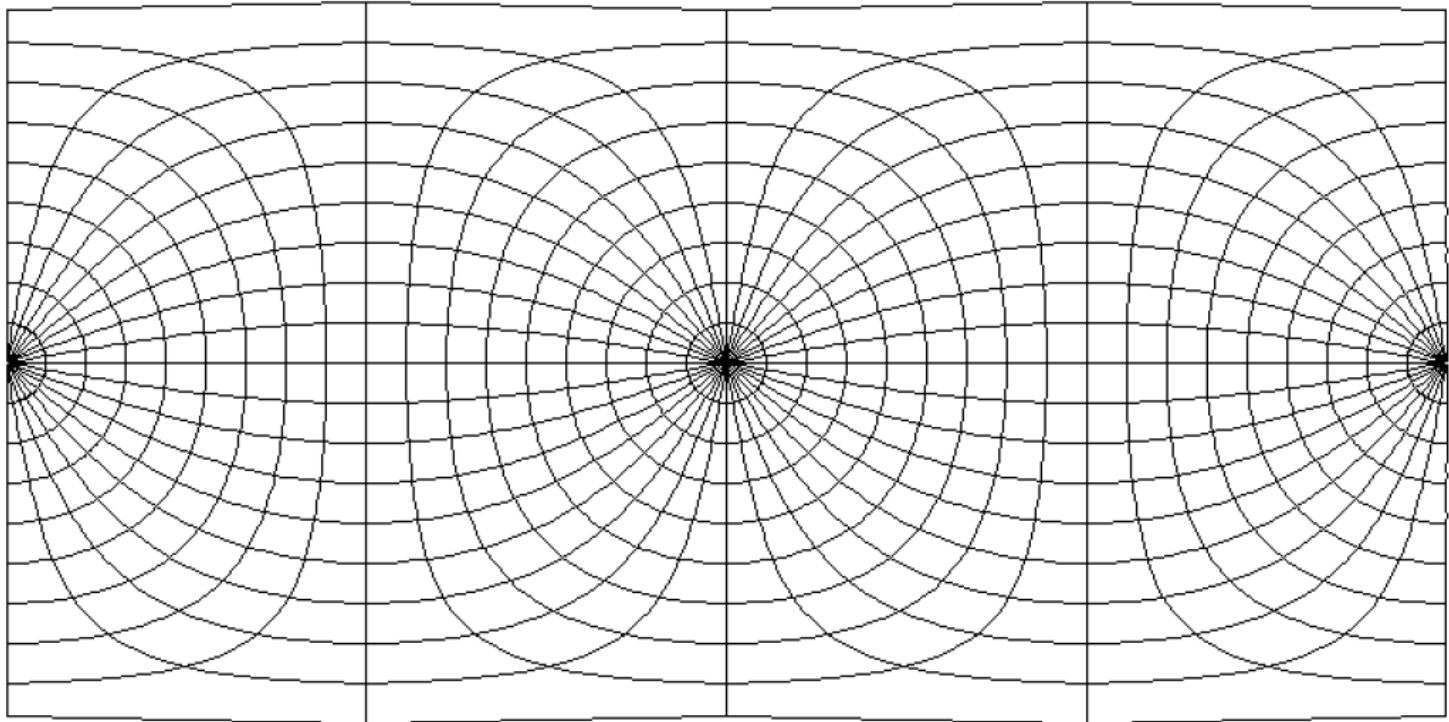
Other projections

- **Polyconical**
 - Each parallel generally has a different radius
- **Polyhedral**
 - Polyhedral
 - Composite forms
 - Meridian belts
- **Unclassified**
 - (special, ambiguous)

Cassini-Soldner projection

- **Cylindrical equidistant projection in transverse position**
- **Used as a basis for maps of the Stable Cadastre in a scale of 1 : 2880**
- **The cylinder touches the reference surface in an undistorted meridian that passes through the center of the territory.**
- **Disadvantage: it is not a conformal display, it distorts the angles**
- **Rapid growth of distortion, a total of 11 coordinate systems: Bohemia = Gusterberg, Moravia = St. Stephen**

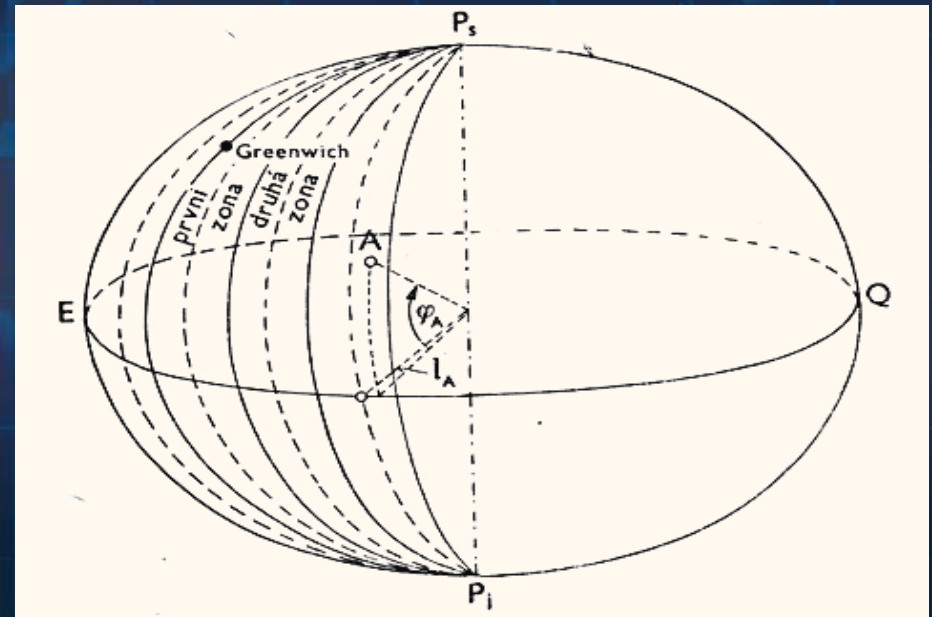
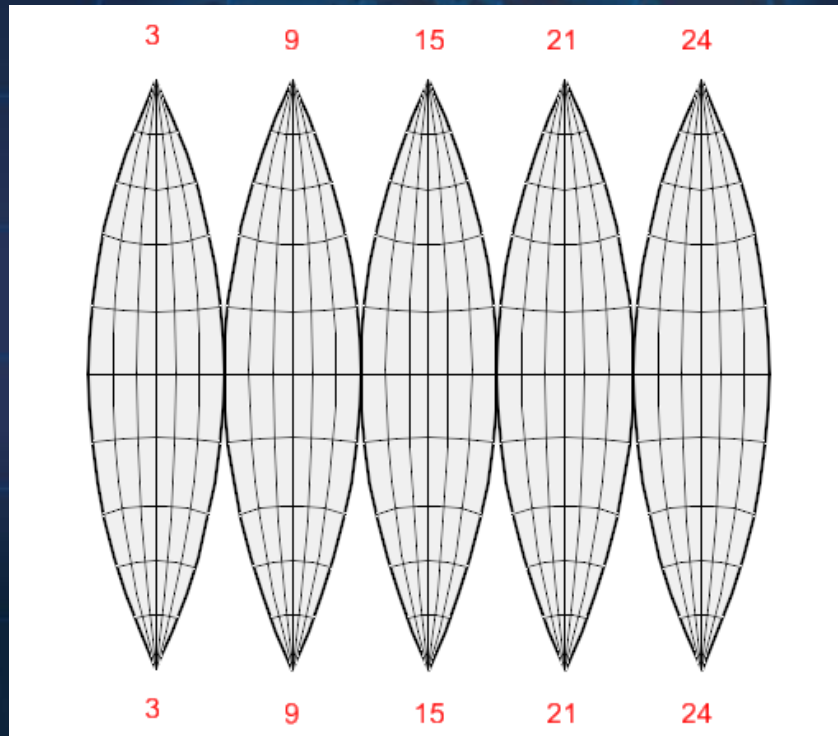
Cassini-Soldner projection



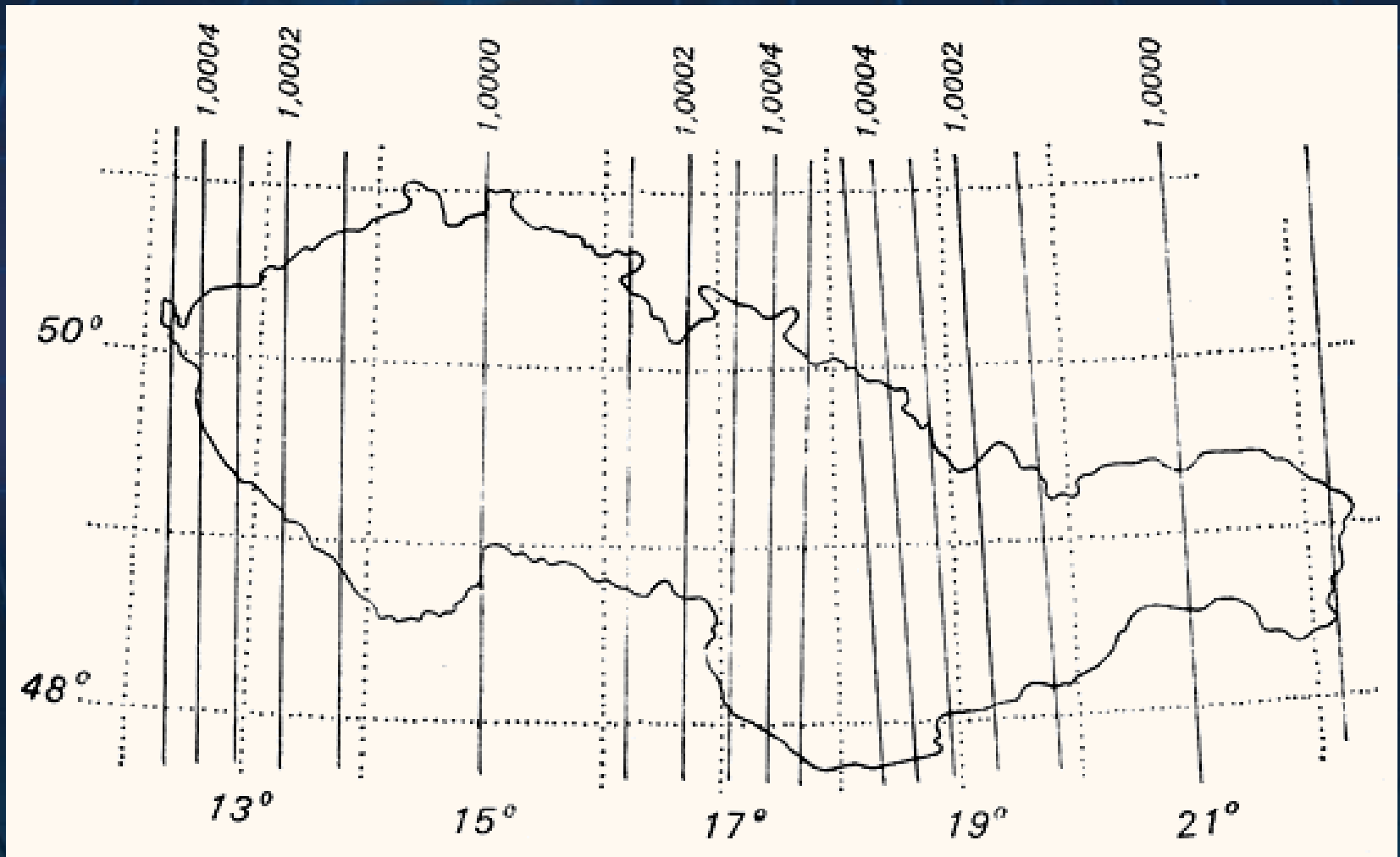
Gauss-Krüger projection

- Direct mapping from ellipsoid to plane, conformal
 - Cylinder in transverse position, undistorted base meridian
 - Territory shown by belts (3° or 6° – Krüger)
 - Each strip separately projected to the plane, has its own coordinate system
-
- In the Czech Republic, used mainly in military maps (6°), also used in civilian maps (3°)
 - System S-52 (Bessel ellipsoid), S-42 (Krasovsky ellipsoid), S-42/83 (Krasovsky ellipsoid)

Gauss-Krüger projection



Gauss-Krüger projection longitudinal distortion

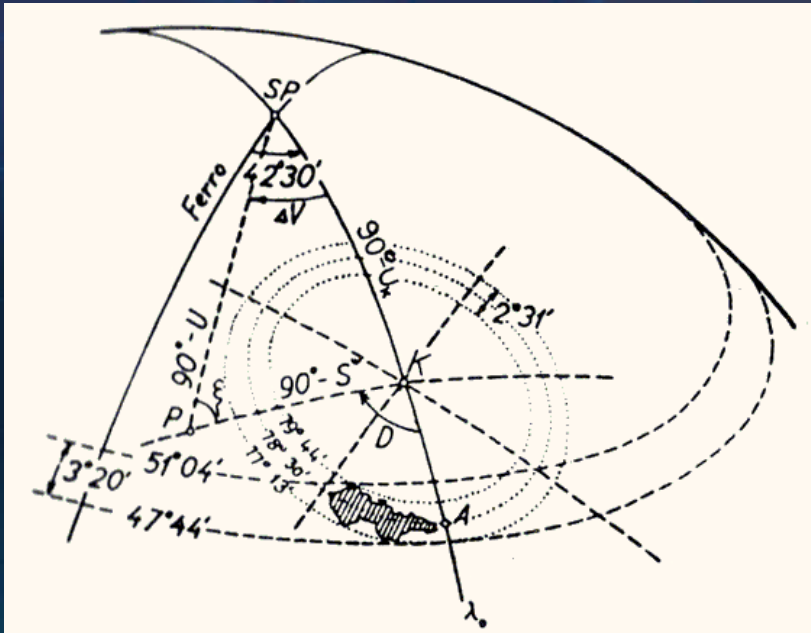


UTM projection

- For military purposes; basic display for NATO
- x axis – Northing , y axis – Easting
- From 80° s.l. to 84° n.l. the projection replaced by UPS (Universal Polar Stereographic projection)
- Used with various ellipsoids:
North America – Clark's el., Europe – Hayford's el.
- Distorted prime meridian ($m = 0.9996$)
- 2 'undistorted' meridians symmetrical with respect to the base meridian.

Křovák's projection

- Double conformal conic representation in oblique position
- Author Ing. Josef Křovák (1922)
- Founded a base for S-JTSK
(system of unified cadastral trigonometric network)



Projection procedure:

1. Conformal representation of an ellipsoid on a sphere (Gaussian)

(Base parallel 49°30')

2. Conformal conic representation in oblique position

(Basic cartographic parallel 78°30'

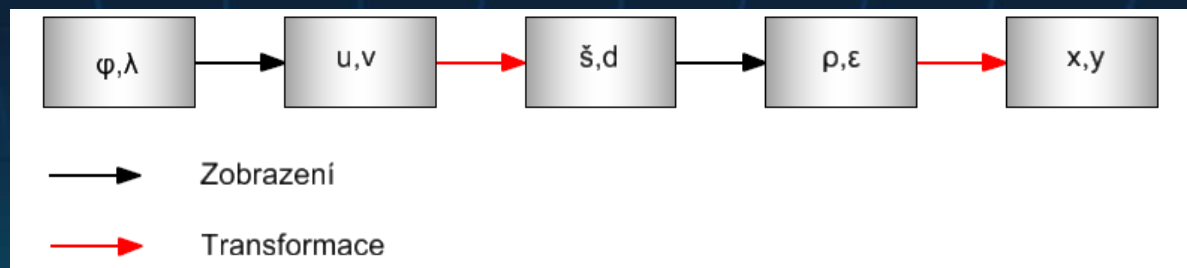
Prime Meridian – Ferro)

Projection derived for one undistorted parallel, but to minimize longitudinal distortions, one of the display equations was multiplied by a factor of 0.9999 and the display thus shows the character of two undistorted parallels

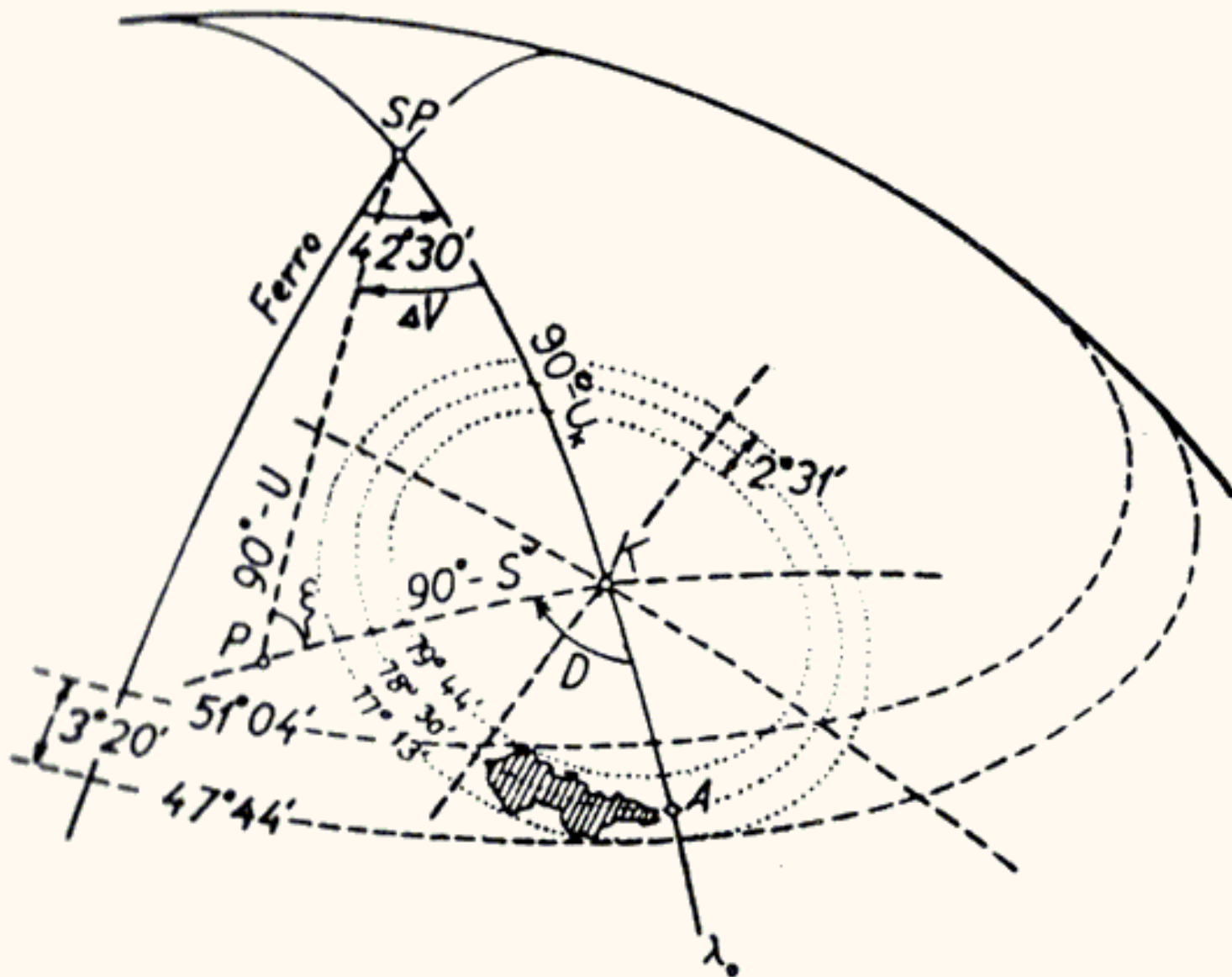
3. In plane:

the X-axis is the image of the meridian 42°30' from Ferro, increasing towards the south

the Y-axis is perpendicular to the X-axis, the origin lies at the top of the cone



Křovák's projection



Křovák's projection – longitudinal distortion

